

IE494, Big Data Processing (Autumn'2025)

Course Placement: B.Tech. (ICT), Technical Elective, Semester – VII

Course format: 3 Lecture hours and 2 Lab hours.

Course Instructor: PM Jat, 2203, Faculty Block 2

Course Content:

Data has taken a different shape in today's time. Modern applications and devices produce data that is very huge, unstructured, and cannot be fitted into the fixed schema. Such data, referred to as "Big Data" does not fit well in conventional storage and processing systems.

A whole new set of techniques and technologies are emerging to process such data. This course teaches various techniques that are used for storing and manipulating "Big Data".

Labs and Assignments:

There would be a series of lab exercises aimed to practice concepts learned in theory. Students will also be required to either do a project or a term paper.

Evaluation plan:

1. Two Mid Semester Exams: 30%
2. Project/Term Paper: 20%
3. Lab Evaluation: 20%
4. End Semester evaluation: 30%

Course Outcomes:

Upon completion of this course, students should be able to

- Explain why traditional data processing techniques do not work on big data problems.
- Work on some of the popular large data processing frameworks like Hadoop, spark, and some NoSQL databases.
- Able to perform various operations like filtering, projecting, aggregation, and OLAP queries on large datasets.
- Appreciate Storage, Indexing, and query execution techniques for big data processing

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x	x		x							x

Detailed Course Contents		
Topic/Content		No. of lectures
1.	Introduction to Big Data: Motivation, Challenges, and Solutions	2
2.	Taste of “Big Data Processing” through Google’s Big-Query	2
3.	Map-Reduce: Technique, Infrastructure, and Programming	3
4.	Spark Architecture, RDD Data model and Programming, Data-frame Data Model Programming, Spark-SQL: infrastructure and programming	4
5.	Concepts of Data Warehousing and OLAP Queries on Spark	4
6.	Overview of NoSQL database systems	3
7.	Data Sharding, Replication, and Replica synchronization techniques for modern distributed databases. Eventual Consistency.	3
8.	One of column family database (Big Table/HBase/Cassandra): Data Model and Programming	4
9.	Column Oriented Storage and Column Oriented Databases: Motivation and evolution of modern column store databases. Various Data Layout and Access Patterns.	4
10.	Implementation of modern column store databases. Vectorized Processing	6
11.	Case Studies	4
Total Lectures		39

References:

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[GFS and Map-Reduce]

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4. White, Tom. “Hadoop: The definitive guide“, 4th ed, O'Reilly Media, Inc., 2015.

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[Spark]

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7. Zaharia, Matei, et al. "Spark: Cluster computing with working sets." *HotCloud* 10.10-10 (2010): 95.
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[Spark-SQL]

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[Data Warehousing and Online Analytical Processing (OLAP)]

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[No-SQL Databases]

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[Consistency in Distributed Databases]

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- 26. Gilbert, Seth, and Nancy Lynch. "Brewer's conjecture and the feasibility of consistent, available, partition-tolerant web services." *Acm Sigact News* 33.2 (2002): 51-59.
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<http://www.bailis.org/blog/linearizability-versus-serializability/>

[Milestones in Big Data Processing]

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[Column Oriented Storage]

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